

FILE : CH12EG1

```
rno=[]
mks=[]
for a in range(4):
    r,m=eval(input("enter roll no.,Marks:"))
    rno.append(r)
    mks.append(m)
d={rno[0]:mks[0],rno[1]:mks[1],rno[2]:mks[2],rno[3]:mks[3]}
print("Creating dictionary")
print(d)
```

#2

```
print()
print("#2")
d={"Vowel1":"a","Vowel2":"e","Vowel3":"i","Vowel4":"o","Vowel5":"u"}
print(d)
print(d["Vowel1"])
```

#3

```
print()
print("#3")
d={"Vowel1":"a","Vowel2":"e","Vowel3":"i","Vowel4":"o","Vowel5":"u"}
print(d["Vowel1"])
print(d)
```

FILE : CH12EG2

```
rno=[]
mks=[]
for a in range(4):
    r,m=eval(input("enter roll no.,Marks:"))
    rno.append(r)
    mks.append(m)
d={rno[0]:mks[0],rno[1]:mks[1],rno[2]:mks[2],rno[3]:mks[3]}
if d[2]>75:
    print("roll number 2 scored",d[2],(">75"))
else:
    print("roll number 2 scored",d[2],("<75"))
print("Creating dictionary")
print(d)
```

FILE : CH12EG3

```
phoneDict={"madhav":1234567,"steven":7654321,"dilpreet":6734521,"
Rabiya":4563217,"Murugan":3241567,"Sampree":4673215}
for a in phoneDict:
    print(a,":",phoneDict[a])
print(phoneDict.keys())
print(phoneDict.values())
if 7654321 in phoneDict.values():
    print("somebody has got that no.")
else:
    print("nobody has got that no.")

if 765432 in phoneDict.values():
    print("somebody has got that no.")
else:
    print("nobody has got that no.")
```

#2

```
print()
print("#2")
dict1={0:"value of key 0",1:"value for key 1","3":"value for a
string -as-a-key",(4,5):"value for a tuple -as-a-key ","and for
fun":7}
print(dict1[0])
print(dict1[(4,5)])
print(dict1[4,5])
print(dict1["3"])
dict1["3"]="change to new string"
print(dict1["3"])
dict1["new"]="new pair is added"
print(dict1["new"])
```

#3

```
print()
print("#3")
birdcount={"finch":10,"Myna":13,"Parakeet":16,"Hornbill":15,"Peac
oke":15}
print(birdcount)
```

FILE : CH12EG4

initiallising of dictionary

```
e={}
e=dict()
e={"name":"Jhon","salary":10000,"age":24}
print(e["name"])
print(e["salary"])
print(e["age"])
print(e)
```

#creating dictionary using zip function

```
e=dict(zip(("name","salary","age"),("Jhon",10000,24)))
print(e)
```

#creating dictionary using a list

```
e=dict([["name","Jhon"],["salary",10000],["age",24]])
print(e)
```

#creating dictionary using a tuple

```
e=dict([("name","Jhon"),("salary",10000),("age",24)])
print(e)
```

#2

#using statement

```
f={"name":"Jhon","salary":10000,"age":24}
del(f["name"])
print(f)
f={"name":"Jhon","salary":10000,"age":24}
del(f)
```

FILE : CH12EG5

```
M={}
n=int(input("How many student?"))
for a in range(n):
    r,m=eval(input("enter Roll no.,Mark:"))
    M[r]=m
print("creating dictionary")
print(M)
```

FILE : CH12EG6

```
M={}
n=int(input("How many student?"))
for a in range(n):
    r,m=eval(input("enter Roll no.,Mark:"))
    M[r]=m
print("creating dictionary")
print(M)
print("to modify")
r=int(input("Enter roll number :"))
if r in M:
    M[r]=float(input("enter NewMarks:"))
else:
    print("No such Roll no. is present")
print("Modify dictionary")
print(M)
```

#2

```
print()
print("#2")
if n in M:
    M[n]=float(input("enter NewMarks:"))
else:
    print("No such Roll no. is present")
print("Modify dictionary")
print(M)
```

#3

```
print()
print("#3")
if m in M:
    M[m]=float(input("enter NewMarks:"))
else:
    print("No such Roll no. is present")
print("Modify dictionary")
print(M)
```

FILE : CH12EG7

```
keys=['one','Two','three']
values=[1,2,3]
numerical=dict(zip(keys,values))
print("Given two list are:",keys,values)
print("Dictionary creating with list is")
print(numerical)
```

#2

```
print()
print("#2")
e={"name":"Jhon","salary":10000,"age":24}
e["dept"]="sales"
print(e)
```

FILE : CH12EG8

```
M={}
n=int(input("How many students?"))
for a in range(n):
    r,m=eval(input("enter Roll No.,Marks :"))
    M[r]=m
print("creating Dictionary")
print(M)
ans='y'
while ans=='y':
    print("Enter detail of New student")
    r,m=eval(input("enter Roll No.,Marks :"))
    M[r]=m
    ans=input("More student ?(y/n):")
print("Dictionary after adding new student")
print(M)
```

FILE : CH12EG8A

```
M={}
n=int(input("How many students?"))
for a in range(n):
    r,m=eval(input("enter Roll No.,Marks :"))
    M[r]=m
print("creating Dictionary")
print(M)
ans=input("More student ?(y/n):")
while ans=='y':
    print("Enter detail of New student")
    r,m=eval(input("enter Roll No.,Marks :"))
    M[r]=m
    ans=input("More student ?(y/n):")
print("Dictionary after adding new student")
print(M)
```

FILE : CH12EG9

```
e={"Jhon":{"age":25,'salary':20000},"Diya":{"age":35,'salary':50000}}
for key in e:
    print("Employee",key,':')
    print('Age:',str(e[key]['age']))
    print('salary:',str(e[key]['salary']))
```

FILE : CH12EG10

```
n=int(input("How many student ?"))
cw={}
for a in range(n):
    keys=input("name of the student :")
    value=int(input("Number of compition won :"))
    cw[keys]=value
print("The dictionary now is :")
print(cw)
```

FILE : CH12EG11

```
M={}
n=int(input("How many students?"))
for a in range(n):
    r,m=eval(input("enter Roll No.,Marks :"))
    M[r]=m
print("creating Dictionary")
print(M)
ans='y'
while ans=='y':
    print("Enter detail of New student")
    r,m=eval(input("enter Roll No.,Marks :"))
    M[r]=m
    ans=input("More student?(y/n):")
print("Dictionary after adding new student")
print(M)
print("Create Dictionary")
print(M)
rno=int(input("roll No. to be deleted? :"))
if rno in M:
    del M[rno]
    print("Roll no.",rno,"delete from dictionary ")
else:
    print("Roll no.",rno,"does not exist in dictionary ")
print("Final dictionary ")
print(M)
```

FILE : CH12EG12

```
import json
s="This is a super ideal This \
idea will change the idea of learning"
word=s.split()
d={}
for one in word:
    key=one
    if key not in d:
        count=word.count(key)
        d[key]=count
print("conting frequencies in list \n",word)
print(json.dumps(d,indent=1))
```

FILE : CH12EG13

```
text="this is sample string"
word=text.split()
print(word)
print("length of text is",len(text))
#2
print()
print("#2")
emp1={'salary':10000,'dept':'Sale','age':24,'name':"Jhon"}
print("the dictionary is",emp1)
print(emp1.get('dept'))
print(emp1.get('desig'))
print(emp1.get('desig',"not yet"))
print("the items(keys) of the dictionary are",emp1.items())
print("the keys of the dictionary are")
for x in emp1:
    print(x)

print("the keys:values of the dictionary are")
for x,y in emp1.items():
    print(x,y)

print("the keys of the dictionary are",emp1.keys())
print("the values of the dictionary are",emp1.values())
```

FILE : CH12EG15

```
l=[]
n=int(input("how many students?"))
for a in range(n):
    r=int(input("enter a roll No. :"))
    l.append(r)
s=dict.fromkeys(l,2500)
print(s)
#s=dict.fromkeys(3) it will give an error
#s=dict.fromkeys(3,) it will give an error
print("created dict.")
print(s)
```

#2

```
print()
print("#2")
nd=dict.fromkeys([2,4,6,8],100)
print(nd)
nd1=dict.fromkeys((3,4,5))
print(nd1)
nd1=dict.fromkeys((3,4,5),(6,7,8))
print(nd1)
```

FILE : CH12EG16

```
new_dict={}
for i in range(10):
    new_dict.setdefault(i)
print("Dictionary is")
print(new_dict)
```

#2

```
print()
print("#2")
emp1={'name':"Jhon",'salary':10000,'age':24}
emp2={'name':"Diya",'salary':54000,'dept':'Sale'}
print("dict 1 : ",emp1)
print("dict 2 : ",emp2)
emp1.update(emp2)
print(emp1)
# almost same as update()
print("# almost same as update()")
for k in emp1.keys():
    emp2[k]=emp1[k]
print(emp2)
# note the difference between update above program snippet
print("# note the difference between update above program snippet
")
```

#3

```
print()
print("#3")
emp2={'name':"Diya",'salary':54000,'dept':'Sale'}
emp=emp2.copy()
print(emp)
#same as copy()
print("#same as copy()")
emp3=emp2
print(emp3)
emp3['age']=24
print(emp3)
```

#4

```
print()
print("#4")
d1={1:[1,2,3],2:[3,4,5]}
d2=d1
print(d2)
d2[1].append(4)
print(d1)
#same as copy()
print("#same as copy()")
d2=d1.copy()
print(d2)
d2[1].append(4)
print(d1)
```

#5

```
print()
print("#5")
old={'name':'vini','age':25}
new=old.copy()
new['name']='Tony'
print(old)
print(new)
```

#6

```
print()
print("#6")
old={'name':['vini','rini'],'age':[25,23]}
new=old.copy()
new['name'].append('Tony')
print(old)
print(new)
```

FILE : CH12EG17

```
stu={1:'Neha',2:'saima',3:'Avnit',4:'Ana',5:'Shaji'}
while len(stu)>=1:
    print("Element deleted :",stu.popitem())
    print(stu)
print("All element of dictionary got deleted in LIFO order ")
print(stu)
```

#2

```
print()
print("#2")
stu={1:'Neha',2:'saima',3:'Avnit',4:'Ana',5:'Shaji'}
print(stu.pop(4))
print("#1",stu)
print(stu.pop(6,"not here"))
print("#2",stu)
print(stu.pop(6,-1))
print("#3",stu)
print(stu.pop(6,None))
print("#4",stu)
stu.clear()
print("#5",stu)
print("note the differene between the above and below")
stu={1:'Neha',2:'saima',3:'Avnit',4:'Ana',5:'Shaji'}
print("#6",stu.clear())
stu={1:'Neha',2:'saima',3:'Avnit',4:'Ana',5:'Shaji'}
print("#7",sorted(stu))
print("#8",sorted(stu,reverse=True))
print("#9",sorted(stu,reverse=False))
```

```

print("same as #7")
print("#10: sorting keys: ",sorted(stu.keys()))
print("#11: sorting values: ",sorted(stu.values()))
print("#12: sorting item: ",sorted(stu.items()))
del(stu)
print("#7",stu)
#3
print()
print("#3")
stu={1: 'Neha',2: 'saima',3: 'Avnit',4: 'Ana',5: 'Shaji'}
key=3
x=stu.pop(key,None)
if x!=None:
    print('Removing pair is',(key,':',x,','))

```

FILE : CH12EG18

```

number={1:111,2:222,3:333,4:444}
print("Giving dictionary is : ",number)
max_key_values=max(number)
min_key_values=min(number)
print("Maximum and Minimum keys:",max_key_values,min_key_values)
max_values=max(number.values())
min_values=min(number.values())
print("Maximum and minimum value:",max_values,min_values)
key_sum=sum(number)
values_sum=sum(number.values())
print("Sum of dictionary's key: ",key_sum)
print("Sum of dictionary's value: ",values_sum)

```

```

#2
print()
print("#2")
d1={(1,2): 'one', (3,4): 'two'}
d2={'Green House':20, 'Blue House':30, 'Gray House':35}
d3={41: 'N',12: 'S',32: 'A',24: 'C'}
d4={1.5: 'N',3.5: 'S',3.2: 'A',2.4: 'C'}
print(min(d1),min(d2),min(d3),min(d4))
print(max(d1),max(d2),max(d3),max(d4))
#print(sum(d1))
print(sum(d3),sum(d4))

```